

Task 1 — Data and Common Sample Window

Five assets analysed:

- SPY
- EUR/USD
- BTC
- ETH
- ATHEX

Data preparation:

- Loaded daily OHLCV data and standardized column names.
- Parsed dates and sorted observations chronologically.
- Combined closing prices into a single five-asset panel.
- Identified the first available date for each asset.
- Aligned all assets to their common sample window.

Key point: The asset with the latest first observation determines the beginning of the common sample.

Task 2 — Risk, Return and Tail Behaviour

Asset	Return	Vol.	Sharpe	Skew.	Kurt.	MDD	VaR95
SPY	14.4%	19.2%	0.66	-0.58	14.04	-33.7%	-1.76%
EURUSD	0.0%	7.4%	-0.24	-0.01	2.30	-23.3%	-0.73%
BTC	28.1%	66.9%	0.39	-0.68	10.80	-83.0%	-6.21%
ETH	24.0%	87.2%	0.25	-0.56	8.52	-93.5%	-7.98%
ATHEX	15.7%	25.7%	0.54	-0.59	11.27	-51.0%	-2.40%

Main findings:

- SPY has the highest Sharpe ratio (0.66), although BTC has the highest raw return (28.1%).
- BTC and ETH have much higher volatility and substantially larger drawdowns.
- SPY has the highest excess kurtosis, while EURUSD has the thinnest tails.
- BTC's daily VaR95 loss magnitude is about $3.5\times$ that of SPY.

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Task 3 — Stylized Facts: ETH and ATHEX

Fact 3 — Unpredictable direction

- Return ACFs are generally close to zero and mostly within the $\pm 1.96/\sqrt{T}$ confidence band.
- This supports weak predictability of return direction.

Fact 4 — Predictable magnitude

- Squared-return ACFs show positive persistence.
- This is evidence of volatility clustering and predictable magnitude.

Fact 5 — Aggregational Gaussianity

Asset	Daily Excess Kurtosis	Weekly Excess Kurtosis
ETH	8.52	3.26
ATHEX	11.27	8.45

- ETH becomes considerably closer to Gaussian at weekly frequency.
- ATHEX also shows a decline, but substantial fat tails remain.

Task 4 — ETH vs. GBM Twin

Method

- Estimated annualized μ and σ from real ETH log returns.
- Generated a same-length GBM price path using the notebook's GBM generator.
- Compared real ETH and the GBM twin using three diagnostics.

Diagnostics

- 1 Return histogram
- 2 Excess kurtosis
- 3 ACF of squared returns

Conclusion

- GBM reproduces the approximate volatility because it is calibrated to the observed data but does not produce fat tails (due to normal returns) or persistent squared - return autocorrelation.
- Therefore, GBM supports Fact 3 as a null model but fails to reproduce Fact 4 and the strong non-Gaussian behaviour of ETH.

Main Observations

My data confirmed several important stylized facts of financial returns.

- Returns showed little autocorrelation, supporting the idea that the direction of returns is difficult to predict.
- Squared returns showed persistence, confirming volatility clustering and predictable magnitude.
- I was surprised by the high excess kurtosis, especially for ATHEX and ETH, showing that extreme movements occur much more often than under a normal distribution.
- Weekly excess kurtosis was lower than daily kurtosis, suggesting that returns become somewhat more Gaussian when aggregated over longer periods.

Overall takeaway

Financial returns are difficult to predict in direction, but their magnitude and risk are not completely random. Fat tails and volatility clustering are important features that simple Gaussian models fail to capture.

Task 5 — The Orchestrator's Memo

Three checks before trusting an AI agent

- ➊ **Data and dates:** Verify first/last observations and check for duplicate dates.
- ➋ **Missing values and observations:** Confirm that the aligned panel contains no missing values and that the number of observations matches the sample period.
- ➌ **Basic statistics:** Independently recalculate annualized volatility and the Sharpe ratio and check that results are consistent.

Two likely failure modes

- Wrong metric definitions or incorrect annualization (e.g. 252 vs. 365 days, simple vs. log returns).
- Data leakage or look-ahead bias, where future information is accidentally used in calculations.

Statistic I would always recompute: Sharpe ratio

The Sharpe ratio combines return and risk into one measure. I would recalculate it using the 2% risk-free rate and check that it agrees with the agent's value within approximately 1%.

Unusually high values, such as above 2 or 3, would deserve further investigation.